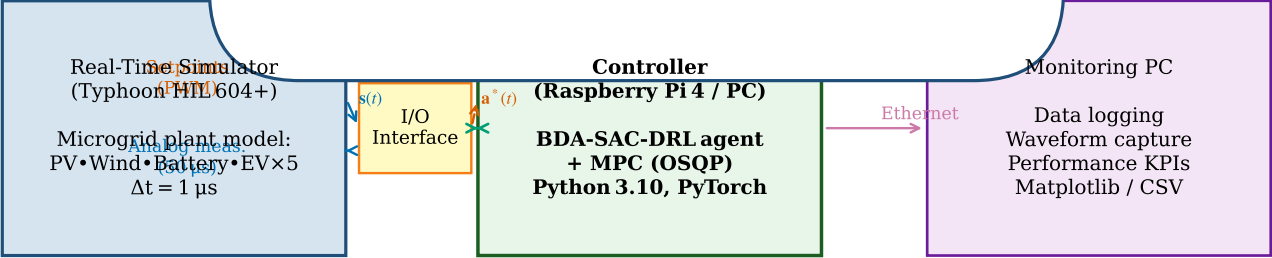


C-HIL Test Bench (Controller Hardware-in-the-Loop)



Test Scenarios

- TC-1: Step-change in PV irradiance (cloud transient)
- TC-2: EV fleet arrival surge (3 EVs within 5 min)
- TC-3: Price spike: RTP jump + \$ { } 0.08 kWh⁻¹ at t=18:00
- TC-4: Intentional SoC boundary violation attempt
- TC-5: Communication delay injection (100 ms, 200 ms)

Validation Metrics

- Tracking RMSE: $|P_{cmd} - P_{act}|$
- SoC boundary violations (count)
- V2G dispatch latency (ms)
- Power setpoint deviation (%)
- Constraint satisfaction rate (%)
- Per IEEE 2030.8-2018